

THE QUESTION

Q.15 · DMG 2019

A fault having both dip-slip as well as strike-slip components is termed as

- a Strike-slip fault
- b Dip-slip fault
- c Oblique-slip fault**
- d Wrench fault



CORRECT ANSWER

Oblique-slip fault — net slip vector has both a dip-slip AND a strike-slip component

TEXTBOOK DEFINITIONS — TWO AUTHORITIES

FOSSEN · STRUCTURAL GEOLOGY · CHAPTER 9 · §9.1 FAULT TERMINOLOGY

"Many faults show some deviation from true dip-slip and strike-slip displacement in the sense that the **net slip vector is oblique**. Such faults are called **oblique-slip faults**. The degree of obliquity is given by the **pitch (also called rake)**, which is the angle between the strike of the slip surface and the slip vector (striation)."

📖 Chapter 9, p. 153 (Fossen, Structural Geology)

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"If the net-slip vector parallels the dip direction of the fault (within $\sim 10^\circ$), the fault is called a **dip-slip fault**; if the vector roughly parallels the strike of the fault, the fault is called a **strike-slip fault**. If the vector is not parallel to either dip direction of the strike, we call the fault an **oblique-slip fault**. As you can see in Figure 8.3, **oblique-slip faults have both a strike-slip and a dip-slip component of movement**."

📖 Page 170 — §8.2, Van der Pluijm & Marshak

FOSSEN · STRUCTURAL GEOLOGY · FIGURE 9.2 CAPTION

"Normal, strike-slip, and reverse faults. **These are end-members of a continuous spectrum of oblique faults**. The spherical projections represent the fault plane (great circle) and the displacement vector."

📖 Figure 9.2, Chapter 9 — Fossen

END-MEMBER 1

Strike-slip fault

Net slip is **horizontal**, parallel to fault strike. No dip-slip component. Vertical or near-vertical fault plane.

Pitch (rake) $\approx 0^\circ$

✔ INTERMEDIATE — THE ANSWER

Oblique-slip fault

Net slip is **oblique** — has BOTH a dip-slip component (up/down) AND a strike-slip component (lateral). The net slip vector is at an intermediate pitch.

Pitch (rake) between 10° and 80°

END-MEMBER 2

Dip-slip fault

Net slip is **in the dip direction**, parallel to the fault dip. No lateral component. Includes normal and reverse faults.

Pitch (rake) $\approx 90^\circ$

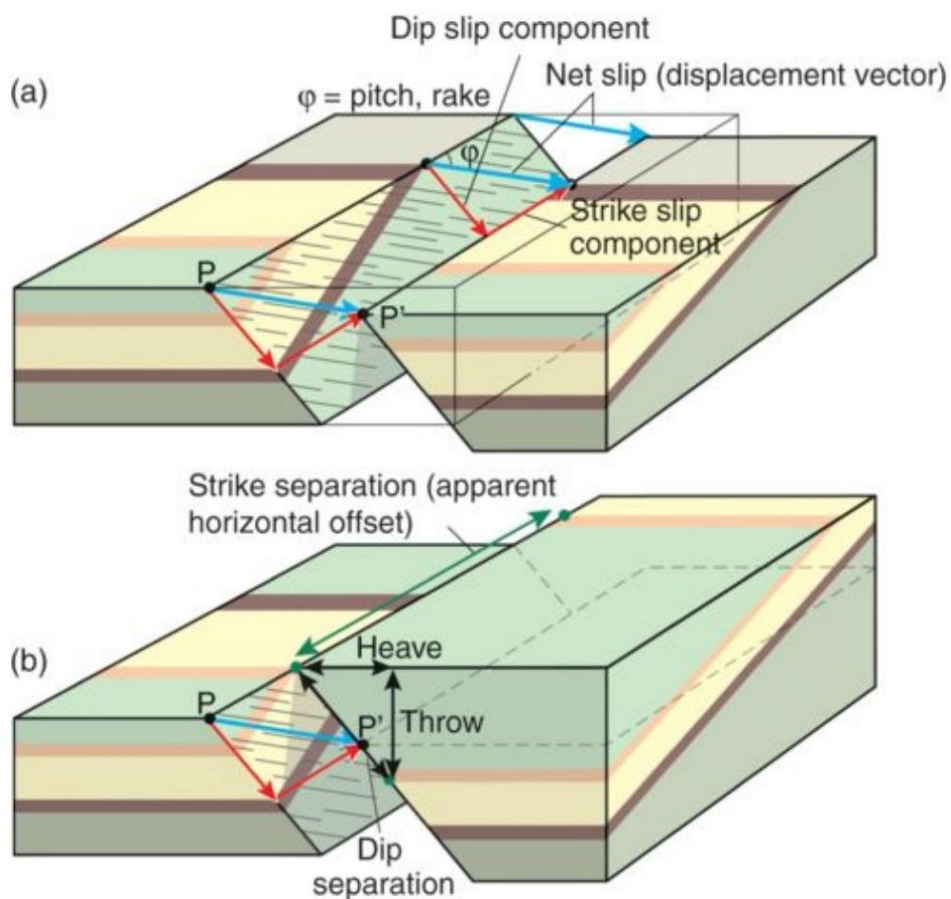


Figure 9.6 Illustration of an oblique-slip fault affecting tilted layers. The fault is normal sinistral and the blue arrows show the true displacement or net slip, which has a dip-slip and a strike-slip component (a). The net slip vector connects points (for example, P and P' in the figure) that were adjacent to each other before faulting. In (b) we have added stratigraphy to the hanging wall up to the level of the footwall. Now the strike and dip separation of the layers, which depend on the orientation of the layers and the nature of the fault displacement,

FIGURE 9.6 (Fossen, Ch. 9) — Illustration of an oblique-slip fault affecting tilted layers. The fault is **normal sinistral** — an oblique-slip type with both a **dip-slip component** (hanging wall down = normal) and a **strike-slip component** (sinistral = left-lateral). The blue arrows show the **net slip (displacement vector)**, which is neither purely dip-parallel nor purely strike-parallel. Note ϕ = pitch (rake) angle between the strike and the slip direction.

SOURCE: FOSSEN — STRUCTURAL GEOLOGY · CHAPTER 9 · FIGURE 9.6

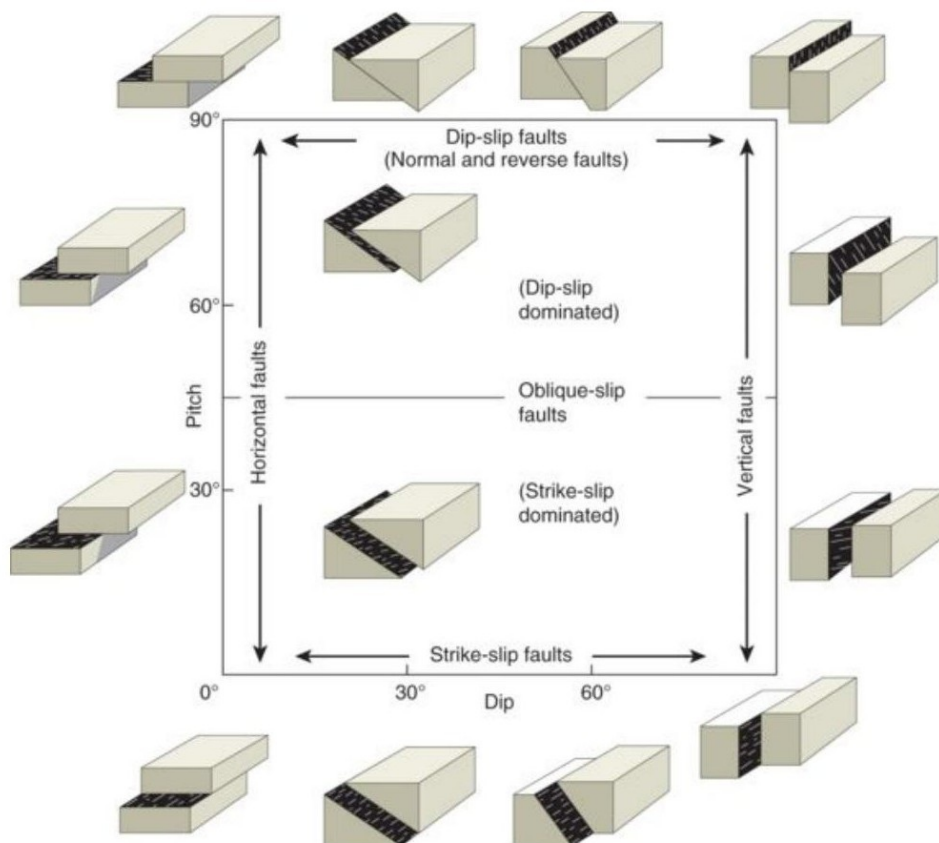


Figure 9.8 Classification of faults based on the dip of the fault plane and the pitch, which is the angle between the slip direction (displacement vector) and the strike.

Based on Angelier ([1994](#)).

FIGURE 9.8 (Fossen, Ch. 9) — Classification of faults based on **dip of the fault plane (x-axis)** and **pitch/rake (y-axis)**. The central zone (intermediate pitch, 30°–60°) is labelled **Oblique-slip faults** — this zone contains all faults with both dip-slip and strike-slip components. Pure dip-slip faults (normal/reverse) plot at the top (pitch 90°); pure strike-slip faults plot at the bottom (pitch 0°). Based on Angelier (1994).

SOURCE: FOSSEN — STRUCTURAL GEOLOGY · CHAPTER 9 · FIGURE 9.8

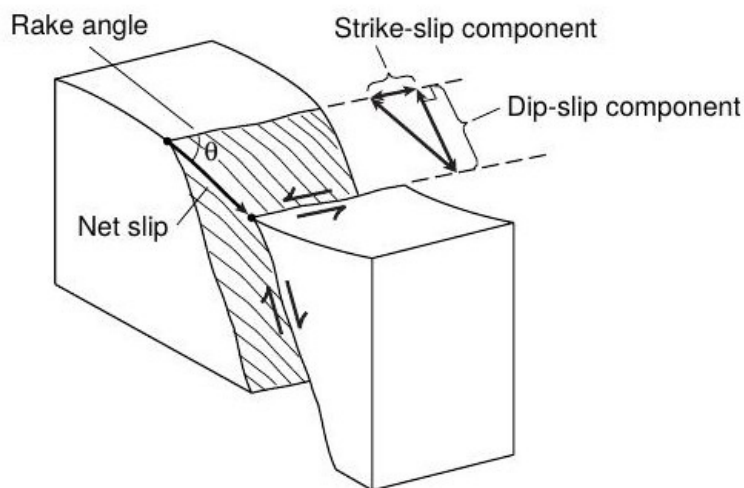


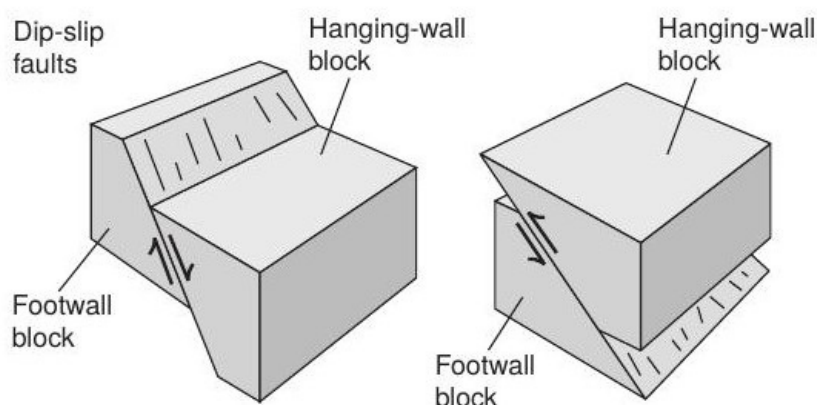
FIGURE 8.3 Block diagram sketch showing the net-slip vector with its strike-slip and dip-slip components, as well as the rake and rake angle.

dip direction, and the **strike-slip component** of net slip in the direction parallel to the strike. If the net-slip vector parallels the dip direction of the fault (within $\sim 10^\circ$), the fault is called a **dip-slip fault**; if the vector roughly parallels the strike of the fault, the fault is called a **strike-slip fault**. If the vector is not parallel to either dip direction of the strike, we call the fault an

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FIGURE 8.3 (VdP & Marshak, p. 169) — Block diagram showing the **net-slip vector** with its **strike-slip component** and **dip-slip component**, as well as the rake angle. When both components exist simultaneously $\rightarrow$ oblique-slip fault.

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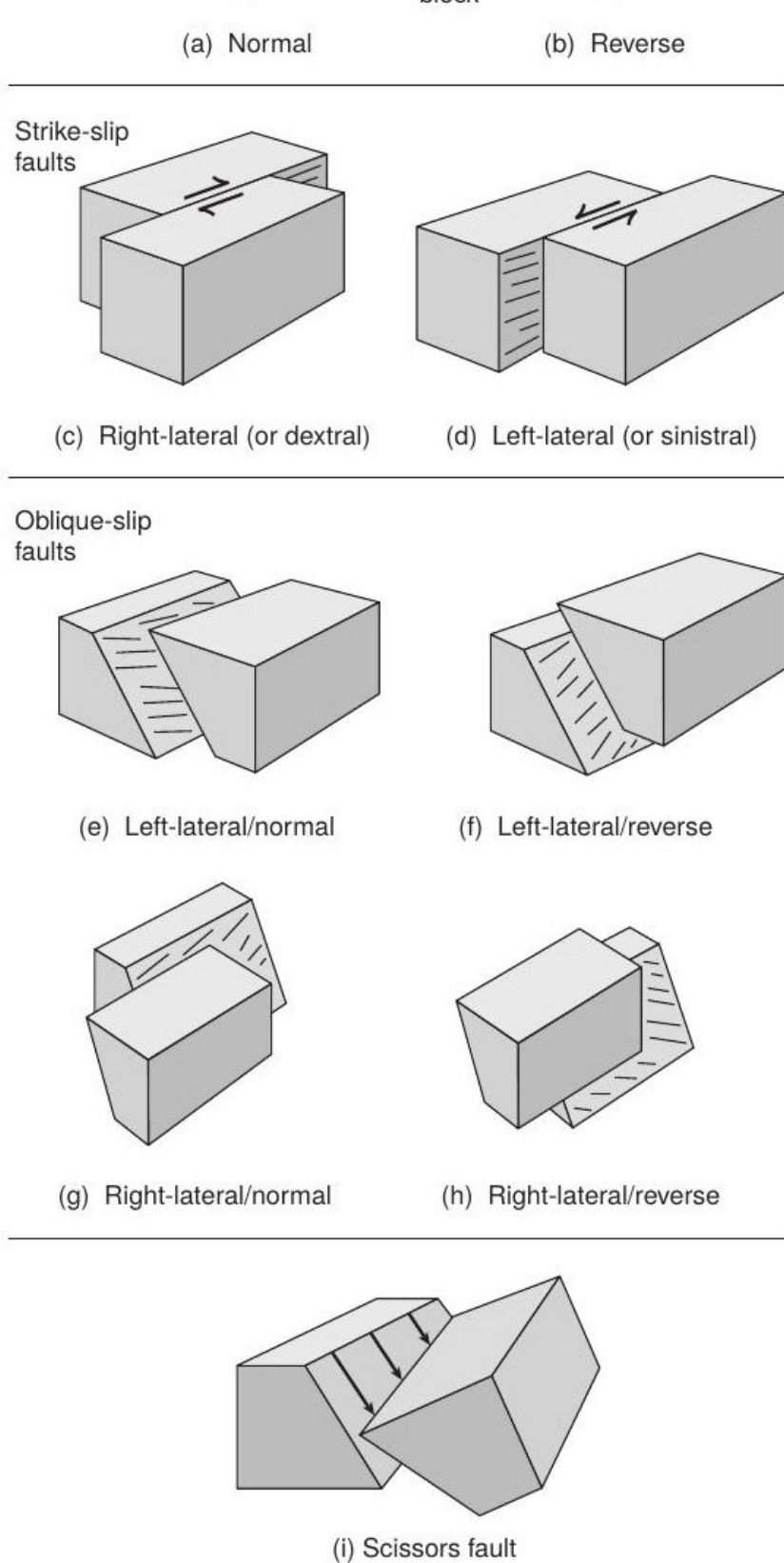


FIGURE 8.4 Block diagram sketches showing the different types of faults.

FIGURE 8.4 (VdP & Marshak, p. 170) — Block diagrams of all fault types. Note (e)–(h) = **oblique-slip faults**: left-lateral/normal, left-lateral/reverse, right-lateral/normal, right-lateral/reverse. These combine dip-slip and strike-slip in one movement.

SUBTYPES OF OBLIQUE-SLIP FAULTS

An oblique-slip fault is described by naming **both components** — the dip-slip sense first, then the strike-slip sense (or vice versa). Four combinations exist:

- Left-lateral / Normal (Sinistral-Normal)**

Hanging wall moves down (normal) AND to the left (sinistral). Example: Figure 9.6 in Fossen.
- Left-lateral / Reverse (Sinistral-Reverse)**

Hanging wall moves up (reverse) AND to the left (sinistral). Common in oblique collision settings.
- Right-lateral / Normal (Dextral-Normal)**

Hanging wall moves down (normal) AND to the right (dextral). Common in pull-apart or transtensional settings.
- Right-lateral / Reverse (Dextral-Reverse)**

Hanging wall moves up (reverse) AND to the right (dextral). Common in oblique convergence / transpressional settings.

ALL FOUR OPTIONS — CONCEPT CARDS

- ✓ **Oblique-slip fault**

Net slip vector is **oblique** — has BOTH a dip-slip component and a strike-slip component. Pitch is between ~10° and ~80°. Fossen: "end-members of a continuous spectrum of oblique faults." Four subtypes (sinistral/dextral × normal/reverse).
- ✗ **Strike-slip fault**

Net slip is **parallel to fault strike** — purely horizontal, no dip-slip component. Pitch ≈ 0°. Can be sinistral (left-lateral) or dextral (right-lateral). This is an *end-member* of the fault spectrum, not a combination.
- ✗ **Dip-slip fault**

Net slip is **parallel to fault dip** — purely in dip direction, no strike-slip component. Pitch ≈ 90°. Includes normal faults (hanging wall down) and reverse faults (hanging wall up). Again, a pure end-member.
- ✗ **Wrench fault**

An **old/informal term for a strike-slip fault** in the petroleum literature. VdP & Marshak (Table 19.1): "A traditional synonym for a strike-slip fault. We *abandon* the term, as it tends to be used inconsistently." A wrench fault is purely strike-slip — no dip-slip component.

SUMMARY COMPARISON TABLE

FAULT TYPE	NET SLIP DIRECTION	DIP-SLIP COMPONENT?	STRIKE-SLIP COMPONENT?	PITCH (RAKE)
Strike-slip fault	Horizontal (parallel to strike)	No	Yes (only)	≈ 0°
Dip-slip fault (normal/reverse)	Down/up the dip direction	Yes (only)	No	≈ 90°

Oblique-slip fault ✓	Oblique (intermediate)	Yes	Yes	10°-80°
Wrench fault	Horizontal (same as strike-slip)	No	Yes (only)	≈ 0°

OPTION-BY-OPTION ANALYSIS

a ✗ INCORRECT

"Strike-slip fault"

A strike-slip fault has movement **only in the horizontal direction**, parallel to the fault's strike. There is no dip-slip component. It is a pure end-member of the fault classification spectrum. Both Fossen (Ch. 9) and Van der Pluijm & Marshak (§8.2, p. 169) explicitly define it as having slip "in the horizontal plane." The question asks for a fault with *both* components — strike-slip alone fails this condition.

→ Strike-slip fault: pitch ≈ 0° — one component only (Fossen Fig. 9.8)

b ✗ INCORRECT

"Dip-slip fault"

A dip-slip fault has movement **only in the dip direction** of the fault surface, resulting in either uplift or subsidence of the hanging wall. There is no lateral (strike-slip) component. Normal faults and reverse faults are the two sub-types. Pitch ≈ 90°. Again, this is a pure end-member — it has only one component of slip, not both.

→ Dip-slip fault: pitch ≈ 90° — one component only (VdP §8.2, p. 169)

c ✓ CORRECT ANSWER

"Oblique-slip fault"

Fossen (Ch. 9): "*Many faults show some deviation from true dip-slip and strike-slip displacement in the sense that the net slip vector is oblique. **Such faults are called oblique-slip faults.***"

Van der Pluijm & Marshak (p. 170): "*If the vector is not parallel to either dip direction of the strike, we call the fault an oblique-slip fault. Oblique-slip faults have **both a strike-slip and a dip-slip component of movement.***"

Both books confirm oblique-slip is the correct term for a fault with both components simultaneously. The slip vector is oblique on the fault surface, giving an intermediate pitch between 0° (pure strike-slip) and 90° (pure dip-slip). Figure 9.8 (Fossen) visually places oblique-slip faults in the center of the classification diagram.

d ✗ INCORRECT — ABANDONED TERM

"Wrench fault"

Wrench fault is an **obsolete term from petroleum geology** for a large-scale continental strike-slip fault. Van der Pluijm & Marshak (Table 19.1, p. 478) state explicitly: "*A traditional synonym for a strike-slip fault... We abandon the term, as it tends to be used inconsistently.*" Critically, a wrench fault has only a *strike-slip* component — it is not a combination of both strike-slip and dip-slip. It therefore cannot answer the question, even before the question of whether the term should be used at all.

→ Wrench fault = informal/abandoned synonym for **strike-slip only**; not both components

EXAM MEMORY AID

Strike-slip = horizontal only → pitch 0° (end-member)

Dip-slip = vertical only → pitch 90° (end-member)

Oblique-slip = BOTH → pitch 10°-80° (intermediate) ✓

Wrench fault = old term for strike-slip → abandoned; not both

Keyword in question: "**both** dip-slip **as well as** strike-slip" = oblique-slip

PROBABLE EXAM QUESTIONS FROM THIS TOPIC

PROBABLE Q · 1

The angle between the slip direction (striation) and the strike of a fault surface is called

- (a) Rake (pitch)
- (b) Plunge
- (c) Trend
- (d) Dip

✓ **Answer: (a)** — Fossen, Ch. 9 Fig. 9.8; VdP & Marshak, Fig. 8.3, p. 169

PROBABLE Q · 2

Normal and strike-slip faults represent which of the following in terms of fault classification?

- (a) Subtypes of oblique-slip faults
- (b) End-members of a continuous spectrum of oblique faults
- (c) Always mutually exclusive categories
- (d) Types classified by dip angle only

✓ **Answer: (b)** — Fossen, Fig. 9.2 caption: "end-members of a continuous spectrum of oblique faults"

PROBABLE Q · 3

A fault where the hanging wall moves downward AND to the left is described as

- (a) Normal sinistral (left-lateral/normal) oblique-slip fault
- (b) Reverse dextral oblique-slip fault
- (c) Strike-slip fault
- (d) Thrust fault

✓ **Answer: (a)** — VdP & Marshak Fig. 8.4e, p. 170; Fossen Fig. 9.6

PROBABLE Q · 4

The term "wrench fault" is considered problematic because

- (a) It implies both dip-slip and strike-slip components
- (b) It is used inconsistently and is now largely abandoned
- (c) It only applies to oceanic plate boundaries
- (d) It refers to a specific kinematic model for thrust faults

✓ **Answer: (b)** — VdP & Marshak, Table 19.1, p. 478: "We abandon the term, as it tends to be used inconsistently"

PROBABLE Q · 5

A fault with a pitch (rake) of 45° would be classified as

- (a) Pure dip-slip
- (b) Pure strike-slip
- (c) Oblique-slip
- (d) Thrust fault

✓ **Answer: (c)** — Fossen, Fig. 9.8; pitch between 10° and 80° = oblique-slip

PROBABLE Q · 6

The vector that connects two formerly adjacent points on either side of a fault is called

- (a) Slip component
- (b) Net slip (displacement vector)
- (c) Throw
- (d) Heave

✓ **Answer: (b)** — Fossen, Ch. 9, p. 153; VdP & Marshak, p. 168